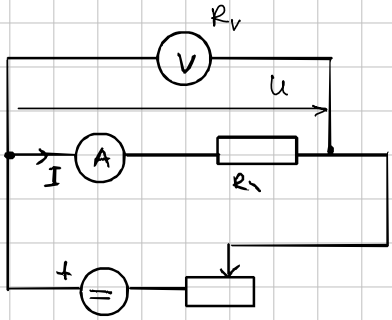
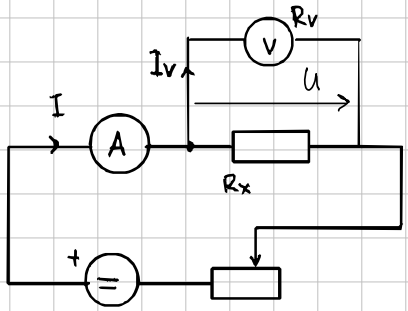
# Mjerenje otpora U-I metodom

a) STRUJNI SPOJ – Za velike otpore, $R_x >> R_A$




$$R_x = \frac{U - I \cdot R_A}{I} = \frac{U}{I} - R_A$$

b) NAPONSKI SPOJ – Za male i srednje otpore $R_x << R_V$




$$R_x = \frac{U}{I - \frac{U}{R_V}}$$

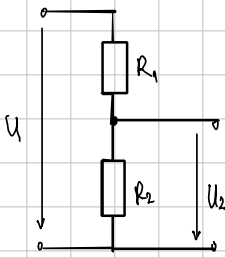
- kod analognih instrumenata
  $R_V = R_K \cdot U_{MO}$
  ↙
  karakterističan otpor [kΩ/V]

- kod digitalnih instrumenata $R_V$ je naveden

## Mjerenje istosmjernog napona

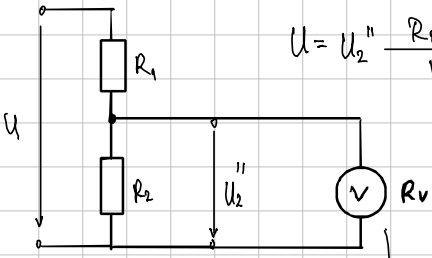
## Mjerenje visokih napona otporničkim djeliteljem

- kad se radi o višim naponima, (do 10 kV) odnosno kad je potrebno prilagoditi amplitudu mjerenog signala ulaznom djelu mjernog instrumenta ili mjernog sustava




$$U_2 = U_1 \cdot \frac{R_2}{R_1 + R_2}$$

$$U = U_2 \frac{R_1 + R_2}{R_2}$$




$$U = U_2'' \frac{R_1 + R_2 \| R_V}{R_2 \| R_V}$$

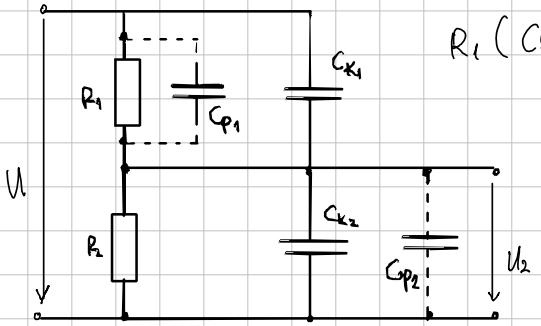
- $R_1, R_2$ = visokog otpora
- disipacija opada povećanjem $(R_1 + R_2)$

neposrednavanje unutrašnjeg otpora voltmetra $R_V$
⇒ korekcija se ne može izračunati

Zanemarivanjem $R_V$ → pogreška mjerenja napona U

# Mjerenje izmjeničnog napona

## Frekvencijska kompenzacija




$$R_1(C_{k1} \parallel C_{p1}) = R_2(C_{k2} \parallel C_{p2})$$

$$C_1 = C_{p1} \parallel C_{k1} \qquad C_2 = C_{k2} \parallel C_{p2}$$

$$\frac{\underline{U_2}}{\underline{U}} = \frac{\underline{Z_2}}{\underline{Z_1} + \underline{Z_2}} = \frac{R_2 \parallel C_2}{R_1 \parallel C_1 + R_2 \parallel C_2}$$

$$\frac{\underline{U_2}}{\underline{U}} = \frac{\dfrac{R_2}{1 + R_2 j\omega C_2}}{\dfrac{R_1}{1 + R_1 j\omega C_1} + \dfrac{R_2}{1 + R_2 j\omega C_2}} = \frac{R_2}{R_1 \dfrac{1 + R_2 j\omega C_2}{1 + R_1 j\omega C_1} + R_2}$$

kada je zadovoljen uvjet $R_1 C_1 = R_2 C_2$

$$\Rightarrow \frac{\underline{U_2}}{\underline{U}} = \frac{R_2}{R_1 + R_2}$$

→ omjer kompleksnih napona nema frekvencijsku ($\omega$) ovisnost ⇒ djelilo je frekvencijski kompenzirano